

User Study – Manual Python Workflow (Method 1 Only)

Instructions: Fill Section A once. For Section B, answer *per use-case*. Duplicate Section B if you completed multiple use-cases.

Participant ID: Amir Your assignment: ☒ Method 1 (Manual Python)
Use case(s) you worked on: ☐ U1 River Flow ☐ U2 Smart-Home Energy ☒ U3 Solar Energy

Section A – Background (answer once)

A1. Educational level (*highest completed*)

☐ High School ☐ Vocational/Technical ☐ Bachelor's ☒ Master's ☐ Doctorate ☐ Other:

A2. Primary role (*pick the closest*)

☐ IoT/CPS Developer ☒ Data Scientist / ML Engineer ☐ Embedded SW Engineer ☐
Educator / Researcher ☐ Software Architect / Full-stack ☐ System Integrator ☐ Other:

A3. Self-rated expertise (*tick one per row*)

	Beginner	Intermediate	Advanced	Expert
Python programming	☐	☐	☒	☐
Time-series / ML	☐	☐	☒	☐
Domain knowledge (your use-case)	☐	☒	☐	☐

A4. Prior experience (*used ML frameworks/libraries before?*) ☒ Yes ☐ No

Section B – Task-Specific (duplicate this page *per use-case*)

This B-section is for: ☐ U1 River Flow ☐ U2 Smart-Home Energy ☒ U3 Solar Energy

B1. Time & Effort

Record your own times. Estimate if needed. “Extra learning time” = time spent reading docs/tutorials.

B1. Time spent (minutes)

	Method 1 (Manual Python)
Writing/testing code (end-to-end)	<u>240</u>
Extra learning time	<u>90</u>

B2. Lines of code (LoC) (*exclude blanks/comments*)

Manual Python LoC: 90

B2. Task Completion & Ease of Use

Completion: None = could not finish; Partial = finished some parts; Complete = all parts.

B3. Were you able to finish the use-case with Manual Python?

Method 1 (Manual Python): ☐ None ☐ Partial ☒ Complete

B4. Ease by workflow step (Manual Python) (1=Very easy, 5=Very difficult; tick one per cell)

Step	1	2	3	4	5
Whole workflow	☐	☐	☐	☒	☐
Data preprocessing	☐	☒	☐	☐	☐
Feature engineering (lags/rolling)	☐	☐	☐	☒	☐
Model training	☐	☐	☐	☒	☐
Model evaluation/plots	☐	☐	☒	☐	☐

B1. Which steps did you complete? (check all that apply)

- ☒ Preprocessing
- ☒ Model configuration
- ☒ Training
- ☒ Evaluation
- ☒ Plotting/Diagnostics

B2. How difficult was each step? (1=Very easy, 5=Very difficult; tick one per row)

Step	1	2	3	4	5
Preprocessing	☐	☒	☐	☐	☐
Model configuration	☐	☐	☐	☒	☐
Training	☐	☐	☐	☒	☐
Evaluation	☐	☐	☒	☐	☐
Plotting/Diagnostics	☐	☐	☒	☐	☐

B3. Notes (issues, decisions, or gaps to revisit):

B5. Time spent per step (record for this use-case; minutes or start/end)

Step	Start (hh:mm)	End (hh:mm)	Minutes
Preprocessing	<u>10:00</u>	<u>10:30</u>	<u>30</u>
Model configuration	<u>10:30</u>	<u>11:30</u>	<u>60</u>
Training	<u>12:00</u>	<u>13:00</u>	<u>60</u>
Evaluation (metrics/baseline)	<u>13:30</u>	<u>14:00</u>	<u>30</u>
Plotting / Diagnostics	<u>14:00</u>	<u>14:30</u>	<u>30</u>
Debugging (errors, crashes, outputs)	<u>15:00</u>	<u>15:30</u>	<u>30</u>
Extra learning / searching / docs	<u>8:30</u>	<u>10:00</u>	<u>90</u>
Total			<u>330</u>

Notes: “Learning” includes reading docs, web searches, tutorials, setup notes, and guides. “Debugging” covers error fixing, crashes, failed runs, and investigating unexpected results.

B6. Comments (bugs, friction points, ideas):

Section E – IoT Use Case Template Reuse (Part 2)

Instructions: This section evaluates your experience reusing the provided IoT pipeline template (e.g., River Flow Forecasting) and adapting it for your own use case. Please answer honestly.

E1. Template Understanding

How clear was the IoT pipeline template (overview, scripts, execution order)?

☐ 1 Very unclear ☐ 2 Unclear ☐ 3 Neutral ☐ 4 Clear ☒ 5 Very clear

E2. Adaptation Effort

How easy was it to adapt the template to your chosen use case?

☐ 1 Very difficult ☐ 2 Difficult ☐ 3 Neutral ☒ 4 Easy ☐ 5 Very easy

E3. Reuse of Offline Training Code

How straightforward was it to reuse the provided offline training code with your dataset?

☐ 1 Not at all ☐ 2 A little ☐ 3 Neutral ☐ 4 Mostly ☒ 5 Very straightforward

Understanding the template: 40
E4. Time & Effort (minutes) Adapting scripts (sensors, gateway, analytics): 60
Running end-to-end pipeline: 30

E5. Problems & Debugging

Did you face errors while reusing the template? ☒ Yes ☐ No

If yes, briefly describe the main issues: In the analytics log console, I was getting a 500 error.

After adjusting sensors.py and correcting the feature field names, the issue was resolved.

How difficult were they to fix?

☐ 1 Very easy ☐ 2 Easy ☒ 3 Neutral ☐ 4 Hard ☐ 5 Very hard

E6. Usefulness

How useful was the template for accelerating your implementation?

☐ 1 Not useful ☐ 2 Slightly useful ☐ 3 Neutral ☐ 4 Useful ☒ 5 Very useful

E7. Suggested Improvements

What changes would make the IoT template easier to reuse for future use cases?

It would be helpful to clearly the required libraries along with their compatible versions,

Scale anchors (for consistency):

- Clarity / Ease: 1 = Very unclear/difficult, 2 = Unclear/difficult, 3 = Neutral, 4 = Clear/easy, 5 = Very clear/easy.
- Debugging difficulty: 1 = Very easy, 5 = Very hard.
- Usefulness: 1 = Not useful, 5 = Very useful.

User Study – ML2++ (Textual DSL)

Instructions: This study is for non-coding users. You will **not** write code or model instances. Use a pre-built ML2++ model and click through the workflow: Save Model → Generate Code → Run Code → Show Forecasting Plots.

Participant ID: Amir Use case(s): ☐ U1 River Flow ☐ U2 Smart-Home Energy
☒ U3 Solar Energy

Section A – Background (answer once)

This section helps us understand your experience level so we can analyze results fairly.

A1. Educational level (*highest completed*)

☐ High School ☐ Vocational/Technical ☐ Bachelor's ☒ Master's ☐ Doctorate ☐ Other: _____

A2. Primary role (*pick the closest*)

☐ IoT/CPS Developer ☐ MDE Practitioner ☒ Data Scientist / ML Engineer ☐ Embedded SW Engineer ☐ Educator / Researcher ☐ Software Architect / Full-stack ☐ System Integrator ☐ Early Adopter / Innovator ☐ Other: _____

A3. Self-rated expertise (*tick one per row*)

	Beginner	Intermediate	Advanced	Expert
IoT/CPS development	☐	☒	☐	☐
Model-Driven Engineering (MDE)	☐	☐	☒	☐
Machine Learning / Data Science	☐	☐	☒	☐

A4. Prior experience (*used DSLs/modeling tools/ML frameworks before?*) ☒ Yes ☐ No

Section B – Task-Specific (duplicate this page *per use case*)

This B-section is for: ☐ U1 River Flow ☐ U2 Smart-Home Energy ☒ U3 Solar Energy

If you did more than one use case, copy this whole Section B for each one and fill it separately.

B1. Task Completion & Ease of Use

B1. Were you able to finish the use case?

☐ None ☐ Partial ☒ Complete

B2. Ease by workflow step (*1=Very easy → 5=Very difficult*)

Step	1	2	3	4	5
Whole workflow (end-to-end)	☐	☐	☒	☐	☐
Data preprocessing (preview/EDA)	☐	☒	☐	☐	☐
Model configuration (preset options)	☐	☐	☐	☒	☐
Run/Execution (logs understanding)	☐	☒	☐	☐	☐
Model evaluation (metrics/plots)	☐	☐	☒	☐	☐

B3. Ease of ML2++ elements (*1=Very easy → 5=Very difficult*)

Element	1	2	3	4	5
Data types & objects	☐	☒	☐	☐	☐
Things & behaviours (statecharts)	☐	☐	☐	☒	☐
Ports & properties	☐	☐	☐	☒	☐
Preset model configuration options	☐	☐	☒	☐	☐
Configuration (instances/connectors)	☐	☐	☒	☐	☐

B2. Learning & Usability**B4. Learning curve** (*1=Very easy → 5=Very difficult*)☐1 ☐2 ☒3 ☐4 ☐5**B5. Intuitiveness** (*1=Very intuitive → 5=Not intuitive*)☐1 ☐2 ☐3 ☒4 ☐5**B6. Hardest part to learn (briefly)**Ports and properties

B3. Debugging & Error Handling**B7. Error frequency** (*0=None, 1=Few, 2=Moderate, 3=Many*)☐0 ☐1 ☒2 ☐3**B8. Clarity of error messages** (*1=Not clear → 5=Very clear*)☐1 ☐2 ☐3 ☒4 ☐5**B9. Debugging effort** (*1=Easy → 5=Hard*)☐1 ☐2 ☒3 ☐4 ☐5**B4. Code Quality, Maintainability, Flexibility***Refers to the generated code you viewed (no editing required).***B10. Generated code quality** (*1=Poor → 5=Excellent*)☐1 ☐2 ☐3 ☒4 ☐5**B11. Maintainability** (*1=Very difficult → 5=Very easy*)☐1 ☐2 ☐3 ☐4 ☒5**B12. Flexibility / Customisation** (*1=None → 5=Very flexible*)☐1 ☐2 ☐3 ☒4 ☐5

Section C – Overall Assessment (ML2++ only)**C1. Overall satisfaction** (*1=Very dissatisfied → 5=Very satisfied*)☐1 ☐2 ☐3 ☒4 ☐5**C2. Would you use ML2++ for your next similar project?**☒ Yes ☐ No ☐ Undecided *Why?* Quickness in changing new requirements

C3. Feature effectiveness (*1=Ineffective → 5=Very effective*)

Feature	1	2	3	4	5
Time-series forecasting models	☐	☐	☐	☒	☐
Automatic preprocessing & plots	☐	☐	☐	☒	☐
Training configuration options	☐	☐	☒	☐	☐
Multi-day forecasting accuracy	☐	☐	☐	☒	☐

C4. Favourite feature: Time series forecasting

C5. Least favourite feature: Training configuration

C6. Missing features you would like: _____

Section D – Open Feedback

D1. Barriers to adoption (*what would stop you from using ML2++ in production?*)

Need to process unstructured datasets (e.g. new)

D2. One improvement you would prioritize

Supporting pipeline definition to bigdata processing

Scale anchors used throughout for consistency:

- Ease / Learning: 1=Very easy, 2=Easy, 3=Moderate, 4=Difficult, 5=Very difficult.
- Intuitiveness: 1=Very intuitive, 2=Intuitive, 3=Neutral, 4=Somewhat confusing, 5=Not intuitive.
- Error clarity: 1=Not clear, 2=Somewhat clear, 3=Clear, 4=Very clear, 5=Extremely clear.
- Code quality: 1=Poor, 2=Fair, 3=Good, 4=Very good, 5=Excellent.
- Maintainability: 1=Very difficult, 2=Difficult, 3=Moderate, 4=Easy, 5=Very easy.
- Flexibility: 1=None, 2=Limited, 3=Some, 4=Flexible, 5=Very flexible.
- Feature effectiveness: 1=Ineffective, 2=Low, 3=Moderate, 4=Effective, 5=Very effective.